For the use of a Registered Medical Practitioner Only

PRESCRIBING INFORMATION

BORTESUN
(Bortezomib Powder for Solution for Injection 3.5 mg/vial)

COMPOSITION

Bortezomib Powder for Solution for Injection 3.5 mg/vial
Each vial contains:
Bortezomib …….3.5 mg
Excipients: Mannitol

DESCRIPTION

BORTESUN is White to off white lyophilized cake or powder in 10 ml colourless tubular glass vial with grey rubber stopper sealed with light green F/O both sides clear lacquer coated aluminium seal.

The reconstituted product should be a clear and colorless solution. If any discoloration or particulate matter is observed, the reconstituted product should not be used.
Unit vial of BORTESUN supply in single carton without Diluents/Reconstitution solution.
BORTESUN reconstituted with 0.9% sodium chloride solution.

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

Pharmacodynamic properties

Mechanism of action

Bortezomib is a reversible inhibitor of the chymotrypsin-like activity of the 26S proteasome in mammalian cells. The 26S proteasome is a large protein complex that degrades ubiquitinated proteins. The ubiquitin-proteasome pathway plays an essential role in regulating the intracellular concentration of specific proteins, thereby maintaining homeostasis within cells. Inhibition of the 26S proteasome prevents this targeted proteolysis which can affect multiple signaling cascades within the cell. This disruption of normal homeostatic mechanisms can lead to cell death. Bortezomib is cytotoxic to a variety of cancer cell types *in vitro*. Bortezomib has been reported to cause a delay in tumor growth *in vivo* in nonclinical tumor models, including multiple myeloma.

Pharmacokinetics

Following intravenous bolus administration of a 1.0 mg/m^2 and 1.3 mg/m^2 dose to patients with multiple myeloma, the mean first-dose maximum plasma concentrations of bortezomib have been reported to be

57 and 112 ng/mL, respectively. In subsequent doses, mean maximum plasma concentrations have been reported to range from 67 to 106 ng/mL for the 1.0 mg/m^2 dose and 89 to 120 ng/mL for the 1.3 mg/m^2 dose. The mean elimination half-life of bortezomib upon multiple dosing has been reported to range from 40 – 193 hours. Bortezomib is eliminated more rapidly following the first dose compared to subsequent doses. Mean total body clearances have been reported to be 102 and 112 L/h following the first dose for doses of 1.0 mg/m^2 and 1.3 mg/m^2, respectively, and ranged from 15 to 32 L/h following subsequent doses for doses of 1.0 mg/m^2 and 1.3 mg/m^2, respectively. Following an IV bolus or subcutaneous (SC) injection of a 1.3 mg/m^2 dose to multiple myeloma patients, the total systemic exposure after repeat dose administration (AUC_{last}) has been reported to be equivalent for SC and IV administration. The C_{max} after SC administration (20.4 ng/mL) has been reported to be lower than IV (223 ng/mL). The AUC_{last} geometric mean ratio has been reported to be 0.99 and 90% confidence intervals were 80.18% - 122.80%.

Distribution

The mean distribution volume of bortezomib has been reported to range from 1659 liters to 3294 liters single- or repeat dose IV administration of 1.0 mg/m^2 or 1.3 mg/m^2 to patients with multiple myeloma. This suggests that bortezomib distributes widely to peripheral tissues. The binding of bortezomib to human plasma proteins has been reported to be 83% over the concentration range of 100 - 1000 ng/mL.

Metabolism

Bortezomib is primarily oxidatively metabolized via cytochrome P450 enzymes, 3A4, 2C19, and 1A2. Bortezomib metabolism by CYP 2D6 and 2C9 enzymes is minor. The major metabolic pathway is deboronation to form two deboronated metabolites that subsequently undergo hydroxylation to several metabolites. Deboronated-bortezomib metabolites are inactive as 26S proteasome inhibitors. Plasma levels of metabolites are reported to be low compared to the parent drug at 10 min and 30 min after dosing.

Elimination

The pathways of elimination of bortezomib have not been reported in humans.

Special Populations

<u>Age, Gender, and Race</u>

The pharmacokinetics of bortezomib have been reported following twice weekly intravenous bolus administration of 1.3 mg/m^2 doses to pediatric patients (2-16 years old) with acute lymphoblastic leukemia (ALL) or acute myeloid leukemia (AML). Clearance of bortezomib has been reported to increase with increasing body surface area (BSA). Geometric mean (%CV) clearance has been reported to be 7.79 (25%) L/hr/m^2, volume of distribution at steady-state was 834 (39%) L/m^2, and the elimination half-life was 100 (44%) hours. After correcting for the BSA effect, other demographics such as age, body weight and sex reportedly did not have clinically significant effects on bortezomib clearance. BSA-normalized clearance of bortezomib in pediatric patients has been reported to be similar to that of adults.

The effects of gender, and race on the pharmacokinetics of bortezomib have not been reported.

Hepatic Impairment

The effect of hepatic impairment on the pharmacokinetics of IV bortezomib has been reported in cancer patients at bortezomib doses ranging from 0.5 to 1.3 mg/m^2. The dose-normalised bortezomib AUC has not been reported to alter with mild hepatic impairment. However, the dose normalized mean AUC values have been reported to increase by approximately 60% in patients with moderate or severe hepatic impairment. A lower starting dose is recommended in patients with moderate or severe hepatic impairment, and those patients should be monitored closely.

Renal Impairment

Exposure of bortezomib (dose-normalized AUC and C_{max}) has been reported to be comparable among patients with various degrees of renal impairment (see DOSE AND METHOD OF ADMINISTRATION).

INDICATIONS

BORTESUN is indicated for the treatment of patients with multiple myeloma.

BORTESUN is indicated for the treatment of patients with mantle cell lymphoma

DOSE AND METHOD OF ADMINISTRATION

Important Dosing Guidelines

BORTESUN is for intravenous or subcutaneous use only. Do not administer BORTESUN by any other route. Because each route of administration has a different reconstituted concentration, use caution when calculating the volume to be administered.

The recommended starting dose of BORTESUN is 1.3 mg/m^2. BORTESUN may be administered intravenously at a concentration of 1 mg/ml, or subcutaneously at a concentration of 2.5 mg/ml. BORTESUN retreatment may be considered for patients with multiple myeloma who had previously responded to treatment with BORTESUN and who have relapsed at least 6 months after completing prior BORTESUN treatment. Treatment may be started at the last tolerated dose. When administered intravenously, administer BORTESUN as a 3 to 5 second bolus intravenous injection.

Dosage in Previously Untreated Multiple Myeloma

BORTESUN is administered in combination with oral melphalan and oral prednisone for 9, six-week treatment cycles as shown in Table below. In Cycles 1 to 4, bortezomib is administered twice weekly (Days 1, 4, 8, 11, 22, 25, 29 and 32). In Cycles 5 to 9, bortezomib is administered once weekly (Days 1, 8, 22 and 29). At least 72 hours should elapse between consecutive doses of bortezomib.

Table: Dosage regimen for patients with previously untreated multiple myeloma

Twice weekly bortezomib (Cycles 1 to 4)						
Week	1	2	3	4	5	6

Bortezomib (1.3 mg/m²)	Day 1	-	-	Day 4	Day 8	Day 11	rest period	Day 22	Day 25	Day 29	Day 32	rest period
Melphalan (9 mg/m²) Prednisone (60 mg/m²)	Day 1	Day 2	Day 3	Day 4	-	-	rest period	-	-	-	-	rest period
Once weekly bortezomib (Cycles 5 to 9 when used in combination with Melphalan and Prednisone)												
Week	1				2		3	4		5		6
Bortezomib (1.3 mg/m²)	Day 1	-	-		Day 8		rest period	Day 22		Day 29		rest period
Melphalan (9 mg/m²) Prednisone (60 mg/m²)	Day 1	Day 2	Day 3	Day 4	-	-	rest period	-	-	-	-	rest period

Dose Modification Guidelines for BORTESUN when given in combination with melphalan and prednisone

Prior to initiating any cycle of therapy with BORTESUN in combination with melphalan and prednisone:

- Platelet count should be at least 70 x 10^9/L and the absolute neutrophil count (ANC) should be at least 1.0 x 10^9/L
- Non-hematological toxicities should have resolved to Grade 1 or baseline

Table: Dose modifications during cycles of combination BORTESUN, melphalan and prednisone

Prednisone Therapy	Toxicity Dose Modification or Delay
Hematological toxicity during a cycle: If prolonged Grade 4 neutropenia or thrombocytopenia, or thrombocytopenia with bleeding is observed in the previous cycle	Consider reduction of the melphalan dose by 25% in the next cycle
If platelet count is not above 30 × 10^9/L or ANC is not above 0.75 x 10^9/L on a bortezomib dosing day (other than day 1)	Withhold bortezomib dose
If several bortezomib doses in consecutive cycles are withheld due to toxicity	Reduce bortezomib dose by 1 dose level (from 1.3 mg/m² to 1 mg/m², or from 1 mg/m² to 0.7 mg/m²)
Grade 3 or higher non-hematological toxicities	Withhold bortezomib therapy until symptoms of toxicity have resolved to Grade 1 or baseline. Then, bortezomib may be reinitiated with one dose level reduction (from 1.3 mg/m² to 1 mg/m², or from 1 mg/m² to 0.7 mg/m²). For bortezomib related neuropathic pain and/or peripheral neuropathy, hold or modify bortezomib.

For information concerning melphalan and prednisone, see manufacturer's prescribing information.

Dose modifications guidelines for peripheral neuropathy are provided.

Posology for previously untreated multiple myeloma patients eligible for haematopoietic stem cell transplantation (induction therapy)

Combination therapy with dexamethasone

BORTESUN 3.5 mg powder for solution for injection is administered via intravenous or subcutaneous injection at the recommended dose of 1.3 mg/m^2 body surface area twice weekly for two weeks on days 1, 4, 8, and 11 in a 21-day treatment cycle. This 3-week period is considered a treatment cycle. At least 72 hours should elapse between consecutive doses of BORTESUN.

Dexamethasone is administered orally at 40 mg on days 1, 2, 3, 4, 8, 9, 10 and 11 of the BORTESUN treatment cycle.

Four treatment cycles of this combination therapy are administered.

Combination therapy with dexamethasone and thalidomide

BORTESUN 3.5 mg powder for solution for injection is administered via intravenous or subcutaneous injection at the recommended dose of 1.3 mg/m^2 body surface area twice weekly for two weeks on days 1, 4, 8, and 11 in a 28-day treatment cycle. This 4-week period is considered a treatment cycle. At least 72 hours should elapse between consecutive doses of BORTESUN.

Dexamethasone is administered orally at 40 mg on days 1, 2, 3, 4, 8, 9, 10 and 11 of the BORTESUN treatment cycle.

Thalidomide is administered orally at 50 mg daily on days 1-14 and if tolerated the dose is increased to 100 mg on days 15-28, and thereafter may be further increased to 200 mg daily from cycle 2 (see Table below).

Four treatment cycles of this combination are administered. It is recommended that patients with at least partial response receive 2 additional cycles.

Table: Posology for BORTESUN combination therapy for patients with previously untreated multiple myeloma eligible for haematopoietic stem cell transplantation

Vc+ Dx	Cycles 1 to 4				
	Week	1	2	3	
	Vc (1.3 mg/m^2)	Day 1, 4	Day 8, 11	Rest Period	
	Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-	
Vc+Dx+T	Cycle 1				
	Week	1	2	3	4
	Vc (1.3 mg/m^2)	Day 1, 4	Day 8, 11	Rest Period	Rest Period
	T 50 mg	Daily	Daily	-	-
	T 100 mg[a]	-	-	Daily	Daily
	Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-	-
	Cycles 2 to 4[b]				
	Vc (1.3 mg/m^2)	Day 1, 4	Day 8, 11	Rest Period	Rest Period
	T 200 mg[a]	Daily	Daily	Daily	Daily
	Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-	-

Vc = Bortezomib; Dx=dexamethasone; T=thalidomide

[a] Thalidomide dose is increased to 100 mg from week 3 of Cycle 1 only if 50 mg is tolerated and to 200 mg from cycle 2 onwards if 100 mg is tolerated.

[b] Up to 6 cycles may be given to patients who achieve at least a partial response after 4 cycles

Dosage adjustments for transplant eligible patients

For BORTESUN dosage adjustments, as described under "Dosage and Dose Modifications for Relapsed Multiple Myeloma and Relapsed Mantle Cell Lymphoma" and "Dose Modifications for Peripheral Neuropathy" should be followed.

In addition, when BORTESUN is given in combination with other chemotherapeutic medicinal products, appropriate dose reductions for these products should be considered in the event of toxicities according to the recommendations in the local prescribing information.

Dosage in previously untreated mantle cell lymphoma

BORTESUN (1.3 mg/m^2) is administered intravenously in combination with intravenous rituximab, cyclophosphamide, doxorubicin and oral prednisone (VcR-CAP) for 6 three-week treatment cycles as shown in table below. BORTESUN is administered first followed by rituximab. BORTESUN is administered twice weekly for two weeks (Days 1, 4, 8, and 11) followed by a ten-day rest period on Days 12 to 21. For patients with a response first documented at cycle 6, two additional VcRCAP cycles are recommended. At least 72 hours should elapse between consecutive doses of BORTESUN.

Table: Dosage regimen for patients with previously untreated mantle cell lymphoma

Twice Weekly BORTESUN(6, Three Week Cycles)*								
Week	1					2		3
BORTESUN (1.3 mg/m^2)	Day 1	-	-	Day 4	-	Day 8	Day 11	Rest period
Rituximab (375 mg/m^2) Cyclophosphamide (750 mg/m^2)	Day 1	-	-			-	-	Rest period
Doxorubicin (50 mg/m^2)								
Prednisone (100 mg/m^2)	Day 1	Day 2	Day 3	Day 4	Day 5	-	-	Rest period

* Dosing may continue for 2 more cycles (for a total of 8 cycles) if response is first seen at cycle 6.

Dose modification guidelines for BORTESUN when given in combination with rituximab, cyclophosphamide, doxorubicin and prednisone

Prior to the first day of each cycle (other than Cycle 1):

- Platelet count should be at least 100 x 10^9/L and absolute neutrophil count (ANC) should be at
- least 1.5 x 10^9/L
- Hemoglobin should be at least 8 g/dL (at least 4.96 mmol/L)
- Non-hematologic toxicity should have recovered to Grade 1 or baseline

Interrupt BORTESUN treatment at the onset of any Grade 3 hematologic or non-hematological toxicities, excluding neuropathy [see WARNINGS AND PRECAUTIONS]. For dose adjustments, see table below.

Table: Dose modifications on days 4, 8, and 11 during cycles of combination BORTESUN, rituximab, cyclophosphamide, doxorubicin and prednisone therapy

Toxicity	Dose modification or delay
Hematological toxicity	
Grade 3 or higher neutropenia, or a platelet count not at or above 25 x 10^9/L	Withhold BORTESUN therapy for up to 2 weeks until the patient has an ANC at or above 0.75 x 10^9/L and a platelet count at or

	above 25 x 10^9/L. • If, after BORTESUN has been withheld, the toxicity does not resolve, discontinue BORTESUN • If toxicity resolves such that the patient has an ANC at or above 0.75 x 10^9/L and a platelet count at or above 25 x 10^9/L, BORTESUN dose should be reduced by 1 dose level (from 1.3 mg/m^2 to 1 mg/m^2, or from 1 mg/m^2 to 0.7 mg/m^2).
Grade 3 or higher non-hematological toxicities	Withhold BORTESUN therapy until symptoms of the toxicity have resolved to Grade 2 or better. Then, BORTESUN may be reinitiated with one dose level reduction (from 1.3 mg/m^2 to 1 mg/m^2, or from 1 mg/m^2 to 0.7 mg/m^2). For BORTESUN-related neuropathic pain and/or peripheral neuropathy, hold or modify BORTESUN.

For information concerning rituximab, cyclophosphamide, doxorubicin and prednisone, see manufacturer's prescribing information.

Dosage and Dose Modifications for Relapsed Multiple Myeloma and Relapsed Mantle Cell Lymphoma

BORTESUN (1.3 mg/m^2/dose) is administered twice weekly for two weeks (Days 1, 4, 8, and 11) followed by a ten-day rest period (Days 12 to 21). For extended therapy of more than eight cycles, BORTESUN may be administered on the standard schedule or, for relapsed multiple myeloma, on a maintenance schedule of once weekly for four weeks (Days 1, 8, 15, and 22) followed by a 13-day rest period (Days 23 to 35). At least 72 hours should elapse between consecutive doses of BORTESUN.

Patients with multiple myeloma who have previously responded to treatment with BORTESUN (either alone or in combination) and who have relapsed at least six months after their prior BORTESUN therapy may be started on BORTESUN at the last tolerated dose. Retreated patients are administered BORTESUN twice weekly (Days 1, 4, 8, and 11) every three weeks for a maximum of eight cycles. At least 72 hours should elapse between consecutive doses of BORTESUN. BORTESUN may be administered either as a single agent or in combination with dexamethasone.

BORTESUN therapy should be withheld at the onset of any Grade 3 non-hematological or Grade 4 hematological toxicities excluding neuropathy as discussed below (see WARNINGS AND PRECAUTIONS). Once the symptoms of the toxicity have resolved, BORTESUN therapy may be reinitiated at a 25% reduced dose (1.3 mg/m^2/dose reduced to 1 mg/m^2/dose; 1 mg/m^2/dose reduced to 0.7 mg/m^2/dose).

For dose modifications guidelines for peripheral neuropathy, see Dose modifications for peripheral neuropathy below.

Dose Modifications for Peripheral Neuropathy

Starting BORTESUN subcutaneously may be considered for patients with pre-existing or at high risk of peripheral neuropathy. Patients with pre-existing severe neuropathy should be treated with BORTESUN only after careful risk-benefit assessment.

Patients experiencing new or worsening peripheral neuropathy during BORTESUN therapy may require a decrease in the dose and/or a less dose-intense schedule.

For dose or schedule modification guidelines for patients who experience BORTESUN related neuropathic pain and/or peripheral neuropathy see table below.

Table: Recommended dose modification for BORTESUN related neuropathic pain and/or peripheral sensory or motor neuropathy

Severity of Peripheral Neuropathy Signs and Symptoms*	Modification of Dose and Regimen
Grade 1 (asymptomatic; loss of deep tendon reflexes or paresthesias) without pain or loss of function	No action
Grade 1 with pain or Grade 2 (moderate symptoms; limiting instrumental Activities of Daily Living (ADL)**)	Reduce BORTESUN to 1 mg/m^2
Grade 2 with pain or Grade 3 (severe symptoms; limiting self-care ADL***)	Withhold BORTESUN therapy until toxicity resolves. When toxicity resolves reinitiate with a reduced dose of BORTESUN at 0.7 mg/m^2 once per week.
Grade 4 (life- threatening consequences; urgent intervention indicated)	Discontinue BORTESUN

*Grading based on NCI Common Toxicity Criteria CTCAE v4.0.
**Instrumental ADL: refers to preparing meals, shopping for groceries or clothes, using telephone, managing money etc.
***Self-care ADL: refers to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden

Special Populations

Patients with Renal Impairment

The pharmacokinetics of bortezomib are not influenced by the degree of renal impairment. Therefore, dosing adjustments of bortezomib are not necessary for patients with renal insufficiency. Since dialysis may reduce bortezomib concentrations, the drug should be administered after the dialysis procedure (see PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES, Pharmacokinetics).

Patients with Hepatic Impairment

Patients with mild hepatic impairment do not require a starting dose adjustment and should be treated per the recommended bortezomib dose. For patients with moderate or severe hepatic impairment, see table below, (also, see PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES, Pharmacokinetics):

Table: Recommended starting dose modification for bortezomib in patients with hepatic impairment

	Bilirubin Level	SGOT (AST) Levels	Modification of Starting Dose in Multiple Myeloma and Relapsed Mantle Cell Lymphoma (1.3 mg/m^2 twice weekly)
Mild	≤ 1.0x ULN	> ULN	None
	> 1.0x − 1.5x ULN	Any	None

Moderate	> 1.5x – 3x ULN	Any	Reduce bortezomib to 0.7 mg/m^2 in the first cycle. Consider dose escalation to 1.0 mg/m^2 or further dose reduction to 0.5 mg/m^2 in subsequent cycles based on patient tolerability.
Severe	> 3x ULN	Any	

Abbreviations: SGOT = serum glutamic oxaloacetic transaminase; AST = aspartate aminotransferase; ULN = upper limit of the normal range.

Method of administration

BORTESUN is administered intravenously or subcutaneously. When administered intravenously, BORTESUN is administered as a 3-5 second bolus intravenous injection through a peripheral or central intravenous catheter followed by a flush with 0.9% sodium chloride solution for injection. For subcutaneous administration, the reconstituted solution is injected into the thighs (right or left) or abdomen (right or left). Injection sites should be rotated for successive injections.

If local injection site reactions occur following BORTESUN injection subcutaneously, a less concentrated BORTESUN solution (1 mg/ml instead of 2.5 mg/ml) may be administered subcutaneously, or changed to IV injection.

CONTRAINDICATIONS

BORTESUN is contraindicated in patients with hypersensitivity to bortezomib, boron or mannitol.

Acute diffuse infiltrative pulmonary and pericardial disease.

When bortezomib is given in combination with other medicinal products, refer to their prescribing information for additional contraindications.

WARNINGS AND PRECAUTIONS

When bortezomib is given in combination with other medicinal products, the prescribing information of these other medicinal products must be consulted prior to initiation of treatment with bortezomib. When thalidomide is used, particular attention to pregnancy testing and prevention requirements is needed.

Bortezomib should be administered under the supervision of a physician experienced in the use of antineoplastic therapy.

There have been fatal cases of inadvertent intrathecal administration of bortezomib. Bortezomib is for IV and subcutaneous use only. DO NOT ADMINISTER BORTEZOMIB INTRATHECALLY.

Overall, the safety profile of patients treated with bortezomib in monotherapy has been reported to be similar to that in patients treated with bortezomib in combination with melphalan and prednisone.

Peripheral Neuropathy

Bortezomib treatment causes a peripheral neuropathy (PN) that is predominantly sensory. However, cases of severe motor neuropathy with or without sensory peripheral neuropathy have been reported. The incidence of peripheral neuropathy increases early in the treatment and has been reported to peak during cycle 5.

Patients with pre-existing symptoms (numbness, pain or a burning feeling in the feet or hands) and/or signs of peripheral neuropathy may experience worsening peripheral neuropathy (including ≥ Grade 3) during treatment with bortezomib. Patients should be monitored for symptoms of neuropathy, such as a burning sensation, hyperesthesia, hypoesthesia, paresthesia, discomfort, neuropathic pain or weakness. The incidence of Grade ≥2 peripheral neuropathy events has been reported to be 24% for SC and 41% for IV ($p = 0.0124$). Grade ≥3 peripheral neuropathy has been reported in 6% for SC compared with 16% for IV ($p=0.0264$). Therefore, patients with pre-existing PN or at high risk of peripheral neuropathy may benefit from starting bortezomib subcutaneously.

Patients experiencing new or worsening peripheral neuropathy may require a change in dose schedule or route of administration to SC (see DOSE AND METHOD OF ADMINISTRATION). Following dose adjustments, improvement in or resolution of peripheral neuropathy has been reported in 51% of multiple myeloma patients with ≥ Grade 2 peripheral neuropathy. Improvement in or resolution of peripheral neuropathy has been reported in 73% of multiple myeloma patients who discontinued due to Grade 2 neuropathy or who had ≥ Grade 3 peripheral neuropathy.

The long-term outcome of peripheral neuropathy has not been reported in mantle cell lymphoma.

Early and regular monitoring for symptoms of treatment-emergent neuropathy with neurological evaluation should be considered in patients receiving bortezomib in combination with medicinal products known to be associated with neuropathy (e.g. thalidomide) and appropriate dose reduction or treatment discontinuation should be considered.

In addition to peripheral neuropathy, there may be a contribution of autonomic neuropathy to some adverse reactions such as postural hypotension and severe constipation with ileus. Information on autonomic neuropathy and its contribution to these undesirable effects is limited.

Hypotension

Bortezomib treatment is commonly associated with orthostatic/postural hypotension. Most adverse reactions are mild to moderate in nature and are reported throughout treatment. Patients who developed orthostatic hypotension on bortezomib (injected intravenously) did not have evidence of orthostatic hypotension prior to treatment with bortezomib. Most patients required treatment for their orthostatic hypotension. Syncopal events have been reported in a minority of patients with orthostatic hypotension. Orthostatic/postural hypotension has not been reported to be acutely related to bolus infusion of bortezomib. The mechanism of this event is unknown although a component may be due to autonomic neuropathy. Autonomic neuropathy may be related to bortezomib or bortezomib may aggravate an underlying condition such as diabetic or amyloidotic neuropathy.

The incidence of hypotension (postural, orthostatic, and Hypotension Not Otherwise Specified) has been reported to be 11% to 12% in multiple myeloma patients. These events have been reported throughout therapy. Caution should be used when treating patients with a history of syncope, patients receiving medications known to be associated with hypotension, and patients who are dehydrated. Management of orthostatic/postural hypotension may include adjustment of antihypertensive medications, hydration, or administration of mineralocorticoids and/or sympathomimetics. Patients should be instructed to seek medical advice if they experience symptoms of dizziness, light-headedness or fainting spells (see UNDESIRABLE EFFECTS).

Cardiac Disorders

Acute development or exacerbation of congestive heart failure, and/or new onset of decreased left ventricular ejection fraction has been reported, including reports in patients with few or no risk factors for decreased left ventricular ejection fraction. Patients with risk factors for, or existing heart disease should be closely monitored. The incidence of any treatment-emergent cardiac disorder has been reported to be 15% and 13% with bortezomib and dexamethasone respectively. The incidence of heart failure events (acute pulmonary edema, cardiac failure, congestive cardiac failure, cardiogenic shock, pulmonary oedema) has been reported to be similar with bortezomib and dexamethasone groups, 5% and 4%, respectively.

There have been isolated cases of QT-interval prolongation; causality has not been reported. Fluid retention may be a predisposing factor for signs and symptoms of heart failure. Patients with risk factors for or existing heart disease should be closely monitored.

Hepatic Events

Rare cases of acute liver failure have been reported in patients receiving multiple concomitant medications and with serious underlying medical conditions. Other reported hepatic events include increases in liver enzymes, hyperbilirubinemia, and hepatitis. Such changes may be reversible upon discontinuation of bortezomib. There is limited re-challenge information in these patients.

Hepatic impairment

Bortezomib is metabolised by liver enzymes. Bortezomib exposure is increased in patients with moderate or severe hepatic impairment; these patients should be treated with bortezomib at reduced doses and closely monitored for toxicities.

Pulmonary Disorders

There have been rare reports of acute diffuse infiltrative pulmonary disease of unknown etiology such as pneumonitis, interstitial pneumonia, lung infiltration and Acute Respiratory Distress Syndrome (ARDS) in patients receiving bortezomib. Some of these events have been fatal. A higher proportion of these events have been reported in Japan. A pre-treatment chest radiograph is recommended to serve as a baseline for potential post-treatment pulmonary changes. In the event of new or worsening pulmonary symptoms, a prompt diagnostic evaluation should be performed and patients treated appropriately.

Death due to ARDS has been reported in patients given high-dose cytarabine (2 g/m^2 per day) by continuous infusion with daunorubicin and bortezomib for relapsed acute myelogenous leukemia in the course of therapy. Therefore, this specific regimen with concomitant administration with high-dose cytarabine (2 g/m^2 per day) by continuous infusion over 24 hours is not recommended.

Laboratory Tests

Complete blood counts (CBC) should be frequently monitored throughout treatment with bortezomib.

Haematological toxicity

Bortezomib treatment is very commonly associated with haematological toxicities (thrombocytopenia, neutropenia and anaemia). Gastrointestinal and intracerebral haemorrhage, have been reported in association with bortezomib treatment. Therefore, platelet counts should be monitored prior to each dose of bortezomib. Bortezomib therapy should be withheld when the platelet count is < 25,000/µl or, in the case of combination with melphalan and prednisone, when the platelet count is ≤ 30,000/µl (see DOSE AND METHOD OF ADMINISTRATION). Potential benefit of the treatment should be carefully weighed against the risks, particularly in case of moderate to severe thrombocytopenia and risk factors for bleeding. Complete blood counts (CBC) with differential and including platelet counts should be frequently monitored throughout treatment with bortezomib. Platelet transfusion should be considered when clinically appropriate

Thrombocytopenia/Neutropenia

Bortezomib treatment is very commonly associated with haematological toxicities (thrombocytopenia, neutropenia and anaemia). Transient thrombocytopenia has been reported in relapsed multiple myeloma and previously untreated mantle cell lymphoma patients treated with bortezomib in combination with rituximab, cyclophosphamide, doxorubicin, and prednisone. Platelets have been reported to be lowest at Day 11 of each cycle of bortezomib treatment and typically recovered to baseline by the next cycle. There was no reported evidence of cumulative thrombocytopenia. The mean platelet count nadir has been reported to be approximately 40% of baseline in patients with multiple myeloma and mantle cell lymphoma. In patients with advanced myeloma the severity of thrombocytopenia has been reported to be related to pre-treatment platelet count: for baseline platelet counts < 75,000/µl, 90% of patients had a count ≤ 25,000/µl, including 14% < 10,000/µl; in contrast, with a baseline platelet count > 75,000/µl, only 14% of patients had a count ≤ 25,000/µl.

A higher incidence (56.7% versus 5.8%) of Grade ≥ 3 thrombocytopenia has been reported in mantle cell lymphoma patients treated with bortezomib, rituximab, cyclophosphamide, doxorubicin and prednisone (VcR-CAP) as compared to patients treated with the rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone (R-CHOP). The overall incidence of all-grade bleeding events has been reported to be similar with both the treatments (6.3% with VcR-CAP and 5.0% with R-CHOP) as well as Grade 3 and higher bleeding events (VcR-CAP: 1.7%; R-CHOP: 1.2%). 22.5% of patients treated with VcR-CAP reportedly received platelet transfusions compared to 2.9% of patients treated with R-CHOP.

Gastrointestinal and intracerebral haemorrhage, have been reported in association with bortezomib treatment. Therefore, platelet counts should be monitored prior to each dose of bortezomib. Bortezomib therapy should be withheld when the platelet count is < 25,000/µl or, in the case of combination with melphalan and prednisone, when the platelet count is ≤ 30,000/µl. Potential benefit of the treatment should be carefully weighed against the risks, particularly in case of moderate to severe thrombocytopenia and risk factors for bleeding.

Complete blood counts (CBC) with differential and including platelet counts should be frequently monitored throughout treatment with bortezomib. Platelet transfusion should be considered when clinically appropriate.

Transient neutropenia that was reversible between cycles has been reported, with no evidence of cumulative neutropenia in patients with mantle cell lymphoma. Neutrophils have been reported to be lowest at Day 11 of each cycle of bortezomib treatment and typically recovered to baseline by the next cycle. Colony stimulating factor support has been reported to be given to 78% of patients treated with VcR-CAP and 61% of patients in treated with R-CHOP. Since patients with neutropenia are at increased risk of infections, they should be monitored for signs and symptoms of infection and treated promptly. Granulocyte colony stimulating factors may be administered for haematologic toxicity according to local standard practice. Prophylactic use of granulocyte colony stimulating factors should be considered in case of repeated delays in cycle administration.

Bortezomib is associated with thrombocytopenia and neutropenia (see UNDESIRABLE EFFECTS). Platelets were lowest at Day 11 of each cycle of bortezomib treatment and typically recovered to baseline by the next cycle. The cyclical pattern of platelet count decrease and recovery has been reported to remain consistent in patients with multiple myeloma and mantle cell lymphoma, with no evidence of cumulative thrombocytopenia or neutropenia.

Platelet counts should be monitored prior to each dose of bortezomib. Bortezomib therapy should be held when the platelet count is <25,000/uL (see DOSE AND METHOD OF ADMINISTRATION and UNDESIRABLE EFFECTS). There have been reports of gastrointestinal and intracerebral hemorrhage in association with bortezomib. Transfusion and supportive care may be considered.

The mean platelet count nadir has been reported to be approximately 40% of baseline in patients with multiple myeloma. The severity of thrombocytopenia related to pretreatment platelet count is shown in Table below. The incidence of significant bleeding events (≥ Grade 3) has been reported to be similar with both bortezomib (4%) and dexamethasone (5%).

Table: Severity of thrombocytopenia related to pretreatment platelet count in multiple myeloma patients treated with bortezomib vs dexamethasone

Pretreatment Platelet Count	Number (%) of patients with platelet count < 10,000/µl	Number (%) of patients with platelet count 10,000-25,000/µl
≥ 75,000/µL	3%	12%
≥ 50,000/µL - < 75,000/µL	14%	79%
≥ 10,000/µL - < 50,000/µL	14%	71%

The incidence of thrombocytopenia adverse events (≥ Grade 4) has been reported to be 32% for bortezomib, rituximab, cyclophosphamide, doxorubicin and prednisone versus 2% for the rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone in patients treated with previously untreated mantle cell lymphoma. The incidence of bleeding adverse events (≥ Grade 3) has been reported to be 1.7% with VcR-CAP and 1.2% with R-CHOP.

No deaths due to bleeding events have been reported with either treatment. No CNS bleeding events have been reported with VcR-CAP treatment; a bleeding event has been reported with R-CHOP. Platelet transfusions have been reported to be given to 23% of the patients treated with VcR-CAP and 3% of the patients treated with R-CHOP.

The incidence of neutropenia (≥ Grade 4) has been reported to be 70% wiht VcR-CAP treatment and 52% with RCHOP treatment. The incidence of febrile neutropenia (≥ Grade 4) has been reported to be 5% in the VcR-CAP treatment and 6% in the R-CHOP treatment. Colony-stimulating factor support has been reported to be provided at a rate of 78% with VcR-CAP treatment and 61% wiht R-CHOP treatment.

Gastrointestinal Adverse Events

Bortezomib treatment can cause nausea, diarrhea, constipation, and vomiting (see UNDESIRABLE EFFECTS) sometimes requiring use of antiemetics and antidiarrheal medications. Fluid and electrolyte replacement should be administered to prevent dehydration. Since patients receiving bortezomib therapy may experience vomiting and/or diarrhea, patients should be advised regarding appropriate measures to avoid dehydration. Patients should be instructed to seek medical advice if they experience symptoms of dizziness, light headedness or fainting spells. Cases of ileus have been uncommonly reported. Therefore, patients who experience constipation should be closely monitored.

Tumor Lysis Syndrome

Because bortezomib is a cytotoxic agent and can rapidly kill malignant cells the complications of tumor lysis syndrome may occur. The patients at risk of tumor lysis syndrome are those with high tumor burden prior to treatment. These patients should be monitored closely and appropriate precautions taken.

Patients with Hepatic Impairment

Bortezomib is metabolized by liver enzymes. Bortezomib exposure is increased in patients with moderate or severe hepatic impairment; these patients should be treated with bortezomib at reduced starting doses and closely monitored for toxicities (see DOSE AND METHOD OF ADMINISTRATION and PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES, Pharmacokinetics).

Posterior Reversible Encephalopathy Syndrome (PRES)

There have been reports of PRES in patients receiving bortezomib. PRES is a rare, reversible, neurological disorder which can present with seizure, hypertension, headache, lethargy, confusion, blindness, and other visual and neurological disturbances. Brain imaging, preferably MRI (Magnetic Resonance Imaging), is used to confirm the diagnosis. In patients developing PRES, discontinue bortezomib. The safety of reinitiating bortezomib therapy in patients previously experiencing PRES is not known.

Herpes zoster virus reactivation

Antiviral prophylaxis is recommended in patients being treated with bortezomib.

The overall incidence of herpes zoster reactivation has been reported to be more common in multiple myeloma patients treated with bortezomib + melphalan + prednisone compared with melphalan + prednisone (14% versus 4% respectively).

The incidence of herpes zoster infection has been reported to be 6.7% in mantle cell lymphoma patients treated with bortezomib, rituximab, cyclophosphamide, doxorubicin and prednisone and 1.2% in patients treated with rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone.

Hepatitis B Virus (HBV) reactivation and infection

When rituximab is used in combination with bortezomib, HBV screening must always be performed in patients at risk of infection with HBV before initiation of treatment. Carriers of hepatitis B and patients with a history of hepatitis B must be closely monitored for clinical and laboratory signs of active HBV infection during and following rituximab combination treatment with bortezomib. Antiviral prophylaxis should be considered. Refer to the prescribing information of rituximab for more information.

Progressive multifocal leukoencephalopathy (PML)

Very rare cases with unknown causality of John Cunningham (JC) virus infection, resulting in PML and death, have been reported in patients treated with bortezomib. Patients diagnosed with PML had prior or concurrent immunosuppressive therapy. Most cases of PML have been reported to be diagnosed within 12 months of their first dose of bortezomib. Patients should be monitored at regular intervals for any new or worsening neurological symptoms or signs that may be suggestive of PML as part of the differential diagnosis of CNS problems. If a diagnosis of PML is suspected, patients should be referred to a specialist in PML and appropriate diagnostic measures for PML should be initiated. Discontinue bortezomib if PML is diagnosed.

Seizures

Seizures have been uncommonly reported in patients without previous history of seizures or epilepsy. Special care is required when treating patients with any risk factors for seizures.

Renal impairment

Renal complications are frequent in patients with multiple myeloma. Patients with renal impairment should be monitored closely.

Concomitant medicinal products

Patients should be closely monitored when given bortezomib in combination with potent CYP3A4-inhibitors. Caution should be exercised when bortezomib is combined with CYP3A4- or CYP2C19 substrates.

Normal liver function should be confirmed and caution should be exercised in patients receiving oral hypoglycemic.

Potentially immunocomplex-mediated reactions

Potentially immunocomplex-mediated reactions, such as serum-sickness-type reaction, polyarthritis with rash and proliferative glomerulonephritis have been reported uncommonly. Bortezomib should be discontinued if serious reactions occur.

Effects on ability to drive and use machines

Bortezomib may cause tiredness, dizziness, fainting, or blurred vision. Patients should be advised not to drive or operate machinery if they experience these symptoms.

DRUG INTERACTIONS

Bortezomib is a weak inhibitor of cytochrome P450 (CYP) isozymes 1A2, 2C9, 2C19, 2D6 and 3A4. Based on the limited contribution (7%) of CYP2D6 to the metabolism of bortezomib, the CYP2D6 poor metabolizer phenotype is not expected to affect the overall disposition of bortezomib.

Bortezomib AUC mean increase of 35% has been reported on co-administration of ketoconazole, a potent CYP3A4 inhibitor and bortezomib. Therefore, patients should be closely monitored when given bortezomib in combination with potent CYP3A4-inhibitors (e.g., ketoconazole, ritonavir).

Omeprazole, a potent inhibitor of CYP2C19 has no significant effect on the pharmacokinetics of bortezomib.

Mean bortezomib AUC reduction of 45% has been reported on co-administration of rifampicin, a potent CYP3A4 inducer and bortezomib. The concomitant use of bortezomib with strong CYP3A4 inducers is therefore not recommended, as efficacy may be reduced. Examples of CYP3A4 inducers are rifampicin, carbamazepine, phenytoin, phenobarbital and St. John’s Wort. Dexamethasone has no significant effect on bortezomib pharmacokinetics.

A 17% increase in mean bortezomib AUC has been reported on co-administration of bortezomib and melphalan-prednisone. This is not considered clinically relevant.

Hypoglycemia and hyperglycemia have been reported in diabetic patients receiving oral hypoglycemics. Patients on oral antidiabetic agents receiving bortezomib treatment may require close monitoring of their blood glucose levels and adjustment of the dose of their antidiabetic medication.

Patients should be cautioned about the use of concomitant medications that may be associated with peripheral neuropathy (such as amiodarone, anti-virals, isoniazid, nitrofurantoin, or statins), or with a decrease in blood pressure.

Drug Laboratory Test Interactions

None known.

UNDESIRABLE EFFECTS

Serious adverse reactions uncommonly reported during treatment with bortezomib include cardiac failure, tumour lysis syndrome, pulmonary hypertension, posterior reversible encephalopathy syndrome, acute diffuse infiltrative pulmonary disorders and rarely autonomic neuropathy.

The most commonly reported adverse reactions during treatment with bortezomib are nausea, diarrhoea, constipation, vomiting, fatigue, pyrexia, thrombocytopenia, anaemia, neutropenia, peripheral neuropathy (including sensory), headache, paraesthesia, decreased appetite, dyspnoea, rash, herpes zoster and myalgia.

Table: Adverse reactions in patients with multiple myeloma treated with bortezomib

System Organ Class	Incidence	Adverse reaction
Infections and infestations	Very common	Nasopharyngitis
	Common	Herpes zoster (inc disseminated & ophthalmic), Pneumonia*, Herpes simplex*, Fungal infection*, pharyngitis, oral candidiasis
	Uncommon	Infection*, Bacterial infections*, Viral infections*, Sepsis (inc septic shock) *, Bronchopneumonia, Herpes virus infection*, Meningoencephalitis herpetic#, Bacteraemia (inc staphylococcal), Hordeolum, Influenza, Cellulitis, Device related infection, Skin infection*, Ear infection*, Staphylococcal infection, Tooth infection*
	Rare	Meningitis (inc bacterial), Epstein-Barr virus infection, Genital herpes, Tonsillitis, Mastoiditis, Post viral fatigue syndrome
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	Rare	Neoplasm malignant, Leukaemia plasmacytic, Renal cell carcinoma, Mass, Mycosis fungoides, Neoplasm benign*
Blood and lymphatic system disorders	Very Common	Thrombocytopenia*, Neutropenia*, Anaemia*
	Common	Leukopenia*, Lymphopenia*
	Uncommon	Pancytopenia*, Febrile neutropenia, Coagulopathy*, Leukocytosis*, Lymphadenopathy, Haemolytic anaemia#
	Rare	Disseminated intravascular coagulation, Thrombocytosis*, Hyperviscosity syndrome, Platelet disorder NOS, Thrombotic microangiopathy (inc thrombocytopenic purpura) #, Blood disorder NOS, Haemorrhagic diathesis, Lymphocytic infiltration
Immune system disorders	Uncommon	Angioedema#, Hypersensitivity*
	Rare	Anaphylactic shock, Amyloidosis, Type III immune complex mediated reaction
Endocrine disorders	Uncommon	Cushing's syndrome*, Hyperthyroidism*, Inappropriate antidiuretic hormone secretion
	Rare	Hypothyroidism

Metabolism and nutrition disorders	Very Common	Decreased appetite
	Common	Dehydration, Hypokalaemia*, Hyponatraemia*, Blood glucose abnormal*, Hypocalcaemia*, Enzyme abnormality*
	Uncommon	Tumour lysis syndrome, Failure to thrive*, Hypomagnesaemia*, Hypophosphataemia*, Hyperkalaemia*, Hypercalcaemia*, Hypernatraemia*, Uric acid abnormal*, Diabetes mellitus*, Fluid retention
	Rare	Hypermagnesaemia*, Acidosis, Electrolyte imbalance*, Fluid overload, Hypochloraemia*, Hypovolaemia, Hyperchloraemia*, Hyperphosphataemia*, Metabolic disorder, Vitamin B complex deficiency, Vitamin B12 deficiency, Gout, Increased appetite, Alcohol intolerance
Psychiatric disorders	Common	Mood disorders and disturbances*, Anxiety disorder*, Sleep disorders and disturbances*
	Uncommon	Mental disorder*, Hallucination*, Psychotic disorder*, Confusion*, Restlessness
	Rare	Suicidal ideation*, Adjustment disorder, Delirium, Libido decreased
Nervous system disorders	Very Common	Neuropathies*, Peripheral sensory neuropathy, Dysaesthesia*, Neuralgia*
	Common	Motor neuropathy*, Loss of consciousness (inc syncope), Dizziness*, Dysgeusia*, Lethargy, Headache*
	Uncommon	Tremor, Peripheral sensorimotor neuropathy, Dyskinesia*, Cerebellar coordination and balance disturbances*, Memory loss (exc dementia) *, Encephalopathy*, Posterior Reversible Encephalopathy Syndrome#, Neurotoxicity, Seizure disorders*, Post herpetic neuralgia, Speech disorder*, Restless legs syndrome, Migraine, Sciatica, Disturbance in attention, Reflexes abnormal*, Parosmia
	Rare	Cerebral haemorrhage*, Haemorrhage intracranial (inc subarachnoid)*, Brain oedema, Transient ischaemic attack, Coma, Autonomic nervous system imbalance, Autonomic neuropathy, Cranial palsy*, Paralysis*, Paresis*, Presyncope, Brain stem syndrome, Cerebrovascular disorder, Nerve root lesion, Psychomotor hyperactivity, Spinal cord compression, Cognitive disorder NOS, Motor dysfunction, Nervous system disorder NOS, Radiculitis, Drooling, Hypotonia, Guillain-Barré syndrome#, Demyelinating polyneuropathy#, ageusia
Eye disorders	Common	Eye swelling*, Vision abnormal*, Conjunctivitis*
	Uncommon	Eye haemorrhage*, Eyelid infection*, Chalazion#, Blepharitis#, Eye inflammation*, Diplopia, Dry eye*, Eye irritation*, Eye pain, Lacrimation increased, Eye discharge
	Rare	Corneal lesion*, Exophthalmos, Retinitis, Scotoma, Eye disorder (inc. eyelid) NOS, Dacryoadenitis acquired, Photophobia, Photopsia, Optic neuropathy#, Different degrees of visual impairment (up to blindness) *
Ear and labyrinth disorders	Common	Vertigo*
	Uncommon	Dysacusis (inc tinnitus) *, Hearing impaired (up to and inc deafness), Ear discomfort*

	Rare	Ear haemorrhage, Vestibular neuronitis, Ear disorder NOS
Cardiac disorders	Uncommon	Cardiac tamponade#, Cardio-pulmonary arrest*, Cardiac fibrillation (inc atrial), Cardiac failure (inc left and right ventricular) *, Arrhythmia*, Tachycardia*, Palpitations, Angina pectoris, Pericarditis (inc pericardial effusion)*, Cardiomyopathy*, Ventricular dysfunction*, Bradycardia,
	Rare	Atrial flutter, Myocardial infarction*, Atrioventricular block*, Cardiovascular disorder (inc cardiogenic shock), Torsade de pointes, Angina unstable, Cardiac valve disorders*, Coronary artery insufficiency, Sinus arrest, New onset of decreased left ventricular ejection fraction
Vascular disorders	Common	Hypotension*, Orthostatic hypotension, Hypertension*
	Uncommon	Cerebrovascular accident#, Deep vein thrombosis*, Haemorrhage*, Thrombophlebitis (inc superficial), Circulatory collapse (inc hypovolaemic shock), Phlebitis, Flushing*, Haematoma (inc perirenal) *, Poor peripheral circulation*, Vasculitis, Hyperaemia (inc ocular) *
	Rare	Peripheral embolism, Lymphoedema, Pallor, Erythromelalgia, Vasodilatation, Vein discolouration, Venous insufficiency
Respiratory, thoracic and mediastinal disorders	Common	Dyspnoea*, Epistaxis, Upper/lower respiratory tract infection*, Cough*, exertional dyspnoea
	Uncommon	Pulmonary embolism, Pleural effusion, Pulmonary oedema (inc acute), Pulmonary alveolar haemorrhage#, Bronchospasm, Chronic obstructive pulmonary disease*, Hypoxaemia*, Respiratory tract congestion*, Hypoxia, Pleurisy*, Hiccups, Rhinorrhoea, Dysphonia, Wheezing
	Rare	Respiratory failure, Acute respiratory distress syndrome, Apnoea, Pneumothorax, Atelectasis, Pulmonary hypertension, Haemoptysis, Hyperventilation, Orthopnoea, Pneumonitis, Respiratory alkalosis, Tachypnoea, Pulmonary fibrosis, Bronchial disorder*, Hypocapnia*, Interstitial lung disease, Lung infiltration, Throat tightness, Dry throat, increased upper airway secretion, Throat irritation, Upper-airway cough syndrome
Gastrointestinal disorders	Very Common	Nausea and vomiting symptoms*, Diarrhoea*, Constipation
	Common	Gastrointestinal haemorrhage (inc mucosal) *, Dyspepsia, Stomatitis*, Abdominal distension, Oropharyngeal pain*, Abdominal pain (inc gastrointestinal and splenic pain) *, Oral disorder*, Flatulence
	Uncommon	Pancreatitis (inc chronic) *, Haematemesis, Lip swelling*, Gastrointestinal obstruction (inc small intestinal obstruction, ileus) *, Abdominal discomfort, Oral ulceration*, Enteritis*, Gastritis*, Gingival bleeding, Gastrooesophageal reflux disease*, Colitis (inc clostridium difficile)*, Colitis ischaemic#, Gastrointestinal inflammation*, Dysphagia, Irritable bowel syndrome, Gastrointestinal disorder NOS, Tongue coated, Gastrointestinal motility disorder*, Salivary gland disorder*
	Rare	Pancreatitis acute, Peritonitis*, Tongue oedema*, Ascites, Oesophagitis, Cheilitis, Faecal incontinence, Anal sphincter atony, Faecaloma*, Gastrointestinal ulceration and

		perforation*, Gingival hypertrophy, Megacolon, Rectal discharge, Oropharyngeal blistering*, Lip pain, Periodontitis, Anal fissure, Change of bowel habit, Proctalgia, Abnormal faeces, eructation, retching, ileus paralytic
Hepatobiliary disorders	Common	Hepatic enzyme abnormality*
	Uncommon	Hepatotoxicity (inc liver disorder), Hepatitis*, Cholestasis
	Rare	Hepatic failure, Hepatomegaly, Budd-Chiari syndrome, Cytomegalovirus hepatitis, Hepatic haemorrhage, Cholelithiasis
Skin and subcutaneous tissue disorders	Common	Rash*, Pruritus*, Erythema, Dry skin
	Uncommon	Erythema multiforme, Urticaria, Acute febrile neutrophilic dermatosis, Toxic skin eruption, Toxic epidermal necrolysis#, Stevens-Johnson syndrome#, Dermatitis*, Hair disorder*, Petechiae, Ecchymosis, Skin lesion, Purpura, Skin mass*, Psoriasis, Hyperhidrosis, Night sweats, Decubitus ulcer#, Acne*, Blister*, Pigmentation disorder*
	Rare	Skin reaction, Jessner's lymphocytic infiltration, Palmar-plantar erythrodysaesthesia syndrome, Haemorrhage subcutaneous, Livedo reticularis, Skin induration, Papule, Photosensitivity reaction, Seborrhoea, Cold sweat, Skin disorder NOS, Erythrosis, Skin ulcer, Nail disorder
Musculoskeletal and connective tissue disorders	Very Common	Musculoskeletal pain*
	Common	Muscle spasms*, Pain in extremity, Muscular weakness
	Uncommon	Muscle twitching, Joint swelling, Arthritis*, Joint stiffness, Myopathies*, Sensation of heaviness
	Rare	Rhabdomyolysis, Temporomandibular joint syndrome, Fistula, Joint effusion, Pain in jaw, Bone disorder, Musculoskeletal and connective tissue infections and inflammations*, Synovial cyst
Renal and urinary disorders	Common	Renal impairment*
	Uncommon	Renal failure acute, Renal failure chronic*, Urinary tract infection*, Urinary tract signs and symptoms*, Haematuria*, Urinary retention, Micturition disorder*, Proteinuria, Azotaemia, Oliguria*, Pollakiuria
	Rare	Bladder irritation
Reproductive system and breast disorders	Uncommon	Vaginal haemorrhage, Genital pain*, Erectile dysfunction,
	Rare	Testicular disorder*, Prostatitis, Breast disorder female, Epididymal tenderness, Epididymitis, Pelvic pain, Vulval ulceration
Congenital, familial and genetic disorders	Rare	Aplasia, Gastrointestinal malformation, Ichthyosis
General disorders and administration site conditions	Very Common	Pyrexia*, Fatigue, Asthenia
	Common	Oedema (inc peripheral), Chills, Pain*, Malaise*
	Uncommon	General physical health deterioration*, Face oedema*, Injection site reaction*, Mucosal disorder*, Chest pain, Gait disturbance, feeling cold, Extravasation*, Catheter related complication*, Change in thirst*, Chest discomfort, Feeling of body temperature change*, Injection site pain*
	Rare	Death (inc sudden), Multi-organ failure, Injection site haemorrhage*, Hernia (inc hiatus)*, Impaired healing*,

		Inflammation, Injection site phlebitis*, Tenderness, Ulcer, Irritability, Non-cardiac chest pain, Catheter site pain, Sensation of foreign body
Investigations	Common	Weight decreased
	Uncommon	Hyperbilirubinaemia*, Protein analyses abnormal*, Weight increased, Blood test abnormal*, C-reactive protein increased
	Rare	Blood gases abnormal*, Electrocardiogram abnormalities (inc QT prolongation) *, International normalised ratio abnormal*, Gastric pH decreased, Platelet aggregation increased, Troponin I increased, Virus identification and serology*, Urine analysis abnormal*
Injury, poisoning and procedural complications	Uncommon	Fall, Contusion
	Rare	Transfusion reaction, Fractures*, Rigors*, Face injury, Joint injury*, Burns, Laceration, Procedural pain, Radiation injuries*
Surgical and medical procedures	Rare	Macrophage activation

NOS=not otherwise specified
* Grouping of more than one MedDRA preferred term.
# Post-marketing adverse reaction regardless of indication

Table: Adverse reactions in patients with mantle cell lymphoma treated with bortezomib with rituximab, cyclophosphamide, doxorubicin and prednisone

System Organ Class	Incidence	Adverse reaction
Infections and infestations	Very Common	Pneumonia*
	Common	Sepsis (inc septic shock) *, Herpes zoster (inc disseminated & ophthalmic), Herpes virus infection*, Bacterial infections*, Upper/lower respiratory tract infection*, Fungal infection*, Herpes simplex*
	Uncommon	Hepatitis B, Infection*, Bronchopneumonia
Blood and lymphatic system disorders	Very Common	Thrombocytopenia*, Febrile neutropenia, Neutropenia*, Leukopenia*, Anaemia*, Lymphopenia*
	Uncommon	Pancytopenia*
Immune system disorders	Common	Hypersensitivity*
	Uncommon	Anaphylactic reaction
Metabolism and nutrition disorders	Very Common	Decreased appetite
	Common	Hypokalaemia*, Blood glucose abnormal*, Hyponatraemia*, Diabetes mellitus*, Fluid retention
	Uncommon	Tumour lysis syndrome
Psychiatric disorders	Common	Sleep disorders and disturbances*
Nervous system disorders	Very Common	Peripheral sensory neuropathy, Dysaesthesia*, Neuralgia*
	Common	Neuropathies*, Motor neuropathy*, Loss of consciousness (inc syncope), Encephalopathy*, Peripheral sensorimotor neuropathy, Dizziness*, Dysgeusia*, Autonomic neuropathy
	Uncommon	Autonomic nervous system imbalance
Eye disorders	Common	Vision abnormal*
Ear and labyrinth disorders	Common	Dysacusis (inc tinnitus) *
	Uncommon	Vertigo*, Hearing impaired (up to and inc deafness)

Cardiac disorders	Common	Cardiac fibrillation (inc atrial), Arrhythmia*, Cardiac failure (inc left and right ventricular) *, Myocardial ischaemia, Ventricular dysfunction*
	Uncommon	Cardiovascular disorder (inc cardiogenic shock)
Vascular disorders	Common	Hypertension*, Hypotension*, Orthostatic hypotension
Respiratory, thoracic and mediastinal disorders	Common	Dyspnoea*, Cough*, Hiccups
	Uncommon	Acute respiratory distress syndrome, Pulmonary embolism, Pneumonitis, Pulmonary hypertension, Pulmonary oedema (inc acute)
Gastrointestinal disorders	Very Common	Nausea and vomiting symptoms*, Diarrhoea*, Stomatitis*, Constipation
	Common	Gastrointestinal haemorrhage (inc mucosal) *, Abdominal distension, Dyspepsia, Oropharyngeal pain*, Gastritis*, Oral ulceration*, Abdominal discomfort, Dysphagia, Gastrointestinal inflammation*, Abdominal pain (inc gastrointestinal and splenic pain) *, Oral disorder*
	Uncommon	Colitis (inc clostridium difficile) *
Hepatobiliary disorders	Common	Hepatotoxicity (inc liver disorder)
	Uncommon	Hepatic failure
Skin and subcutaneous tissue disorders	Very Common	Hair disorder*
	Common	Pruritus*, Dermatitis*, Rash*
Musculoskeletal and connective tissue disorders	Common	Muscle spasms*, Musculoskeletal pain*, Pain in extremity
Renal and urinary disorders	Common	Urinary tract infection*
General disorders and administration site conditions	Very Common	Pyrexia*, Fatigue, Asthenia
	Common	Oedema (inc peripheral), Chills, Injection site reaction*, Malaise*
Investigations	Common	Hyperbilirubinaemia*, Protein analyses abnormal*, Weight decreased, Weight increased

* Grouping of more than one MedDRA preferred term.

USE IN SPECIAL POPULATIONS

Pregnancy

Women of childbearing potential should avoid becoming pregnant while being treated with bortezomib.

There are no reported data for placental transfer of bortezomib. There are no adequate data reported in pregnant women. If bortezomib is used during pregnancy, or if the patient becomes pregnant while receiving this drug, the patient should be apprised of the potential hazard to the fetus.

Patients should be advised to use effective contraceptive measures during and for 3 months following treatment to prevent pregnancy and to avoid breast feeding during treatment with bortezomib.

There are no reported clinical data for bortezomib with regard to exposure during pregnancy. The teratogenic potential of bortezomib has not been fully investigated.

Bortezomib reportedly had no effects on embryonal/foetal development in rats and rabbits at the highest maternally tolerated doses. There are no reported data on the effects of bortezomib on parturition and post-natal development. Bortezomib should not be used during pregnancy unless the clinical condition of the woman requires treatment with bortezomib.

Thalidomide is a known human teratogenic active substance that causes severe life-threatening birth defects. Thalidomide is contraindicated during pregnancy and in women of childbearing potential unless all the conditions of the thalidomide pregnancy prevention programme are met. Patients receiving bortezomib in combination with thalidomide should adhere to the pregnancy prevention programme of thalidomide. Refer to the prescribing information of thalidomide for additional information.

Nursing Mothers

It is not known whether bortezomib is excreted in human milk. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants from bortezomib, women should be advised against breast feeding while being treated with bortezomib.

Fertility

There are no fertility data reported with bortezomib.

OVERDOSE

IV doses (in monkeys and dogs) approximately two to three times the recommended clinical dose on a mg/m^2 basis are reported to be associated with increases in heart rate, decreases in contractility, hypotension and death. The decreased cardiac contractility and hypotension have been reported to respond to acute intervention with positive inotropic or pressor agents. A slight increase in the corrected QT interval has been reported at a lethal dose in dogs.

Overdosage more than twice the recommended dose in patients has been reported to be associated with the acute onset of symptomatic hypotension and thrombocytopenia with fatal outcomes.

There is no known specific antidote for bortezomib overdosage. In the event of an overdosage, the patient's vital signs should be monitored and appropriate supportive care given to maintain blood pressure (such as fluids, pressors, and/or inotropic agents) and body temperature (see WARNINGS AND PRECAUTIONS and DOSE AND METHOD OF ADMINISTRATION).

STORAGE

Store up to 30°C. Retain in Original pack to protect from light.

Keep this medicine out of the sight and reach of children.

Do not use this medicine after the expiry date stated on the vial and the carton after EXP.
Keep the vial in the outer carton in order to protect from light.
This medicine does not require any special temperature storage conditions.
From a microbiological point of view, the reconstituted solution should be used immediately after preparation. If the reconstituted solution is not used immediately, in-use storage times and conditions prior to use are the responsibility of the user. However, the reconstituted solution is stable for 8 hours at 25ºC stored in the original vial and/or a syringe, with a total storage time for the reconstituted medicine not exceeding 8 hours prior to administration.

SUPPLY

Unit vial of Bortesun supply in single carton along with pack insert.

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